

devices. UPnP AV specifications define the interaction model between UPnP AV devices and associated control point applications. UPnP AV devices can include TVs, VCRs, CD/DVD players, set-top boxes, stereo systems, still-image cameras, electronic picture frames, and PCs. The UPnP AV architecture allows devices to support entertainment content in any format and over any transfer protocol. UPnP AV specifications define two types of logical devices on the home network: Media Servers and Media Renderers.

The specifications also define four services hosted by Media Servers and Media Renderers.

- Content Directory Service: Enumerates the available content (videos, music, pictures, etc).
- Connection Manager Service: Determines how the content can be transferred from the Media Server to the Media Renderer devices.
- AV Transport Service: Controls the flow of the content (play, stop, pause, seek, etc.).
- Rendering Control Service: Controls how the content is played (volume/mute, brightness, etc.).

Putting It All Together

The digital home offers significant new opportunities for the CE, mobile, and PC industries. The vision articulated here for digital interoperability will require considerable effort to achieve. The industry needs to align, coordinate, and deliver at several levels as follows:

- **Usages:** The CE, mobile, and PC industries must define and align on a usages roadmap that will drive consumer acceptance of a new category of interoperable digital home products. By necessity, this roadmap will be dynamic and must progressively reflect available technology and standards over time. Digital entertainment and media usages will most likely be the driving factor for early consumer adoption, and the availability of technology and standards dictates a planned evolution from personal to commercial media usages.
- **Interoperability Framework:** CE, mobile, and PC industries must:
 - 1) Align on the framework for digital interoperability,
 - 2) Continue to participate in key standards arenas, such as ISO, the UPnP Forum, CEA and the 1394 Trade Association, to ensure future usages and capabilities are supported, and